Machine Learning CS 4641

Probability and Statistics

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Slides adapted from Max Roozbahani, Le Song , Sam Roweis, and Chao Zhang.

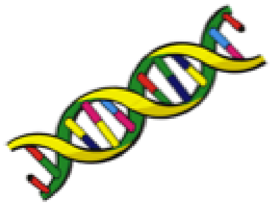
Outline

- Probability Distributions
- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation

Probability

probability of an event = $1/6$
unbiased die

- A sample space S is the set of all possible outcomes of a conceptual or physical, repeatable experiment. (S can be finite or infinite.)
  - E.g., S may be the set of all possible outcomes of a dice roll: S $(1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6)$
  - E.g., S may be the set of all possible nucleotides of a DNA site: S $(A \quad C \quad G \quad T)$



  - E.g., S may be the set of all possible time-space positions of an aircraft on a radar screen.
- An Event A is any subset of S
  - Seeing "1" or "6" in a dice roll; observing a "G" at a site; UA007 in space-time interval

Three Key Ingredients in Probability Theory

A **sample space** is a collection of all possible **outcomes**

Random variables X represents **outcomes** in sample space

Probability of a random variable to happen $p(x) = p(X = x)$

random variable

a particular outcome

probability that r.v. outcome x

$p(x) \geq 0$

pdf & pmf are probability distribution function = PDF

Continuous variable

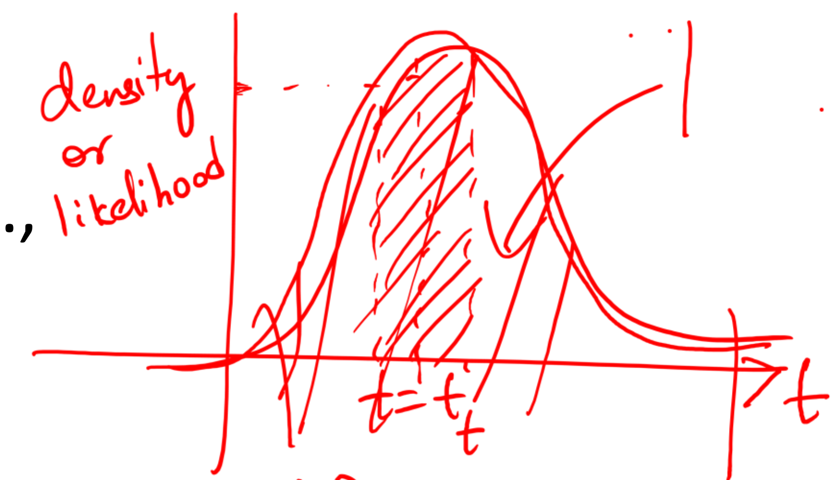
$$\boxed{\int_x p(x)dx = 1}$$

Definition: Takes values from a continuous range (e.g., any real number within an interval).

Distribution: Governed by a **probability density function** (PDF).

Key Concepts:

- The **density** represents likelihood, but not actual probability at a specific point.
- Example: **Temperature**, which can be any real value (e.g., 72.3°F).
- Common distribution: **Gaussian (Normal) Distribution**.



$P(t = t_t) = 0$

Discrete variable

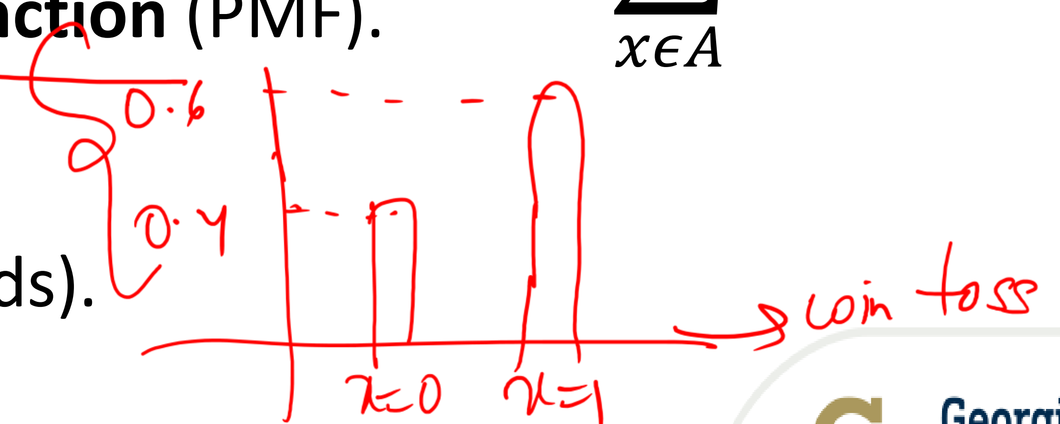
$$\sum_{x \epsilon A} p(x) = 1$$

Definition: Takes values from a countable set (e.g., integers).

Distribution: Governed by a **probability mass function** (PMF).

Key Concepts:

- The function directly gives **probability values**.
- Example: **A coin flip** (e.g., 0 for tails, 1 for heads).
- Common distribution: **Bernoulli Distribution**.



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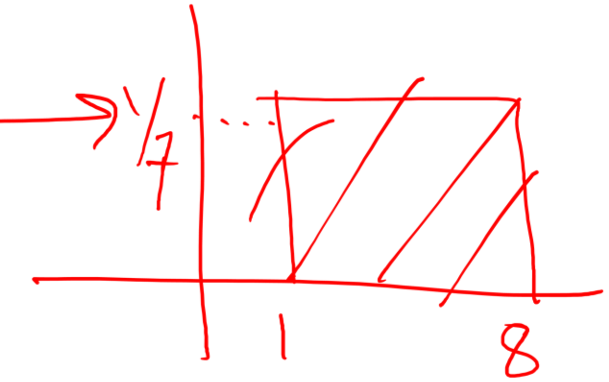
is to optimize objective function $\rightarrow$ parameters θ

training $\approx$ optimization

Continuous Probability Functions



- Examples:
  - Uniform Density Function: $(a, b \in \theta)$

$$\underbrace{f_x(x)}_{\text{objective function}} = \begin{cases} \dfrac{1}{b-a} & for\ a \le x \le b \\ 0 & otherwise \end{cases}$$

$\frac{1}{c-d}$ for $a \le x \le b$

$c = b$
$d = a$

  - Exponential Density Function:

$$f_x(x) = \frac{1}{\mu} e^{-\frac{x}{\mu}} \qquad for\ x \ge 0$$

$\mu \in \theta$.

In our class, μ means expected value or weighted average AND arithmetic mean

  - Gaussian(Normal) Density Function

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\mu, \sigma \in \theta$

$$\frac{1}{\sqrt{2\pi b}} e^{-\frac{(x-a)^2}{2b}}$$

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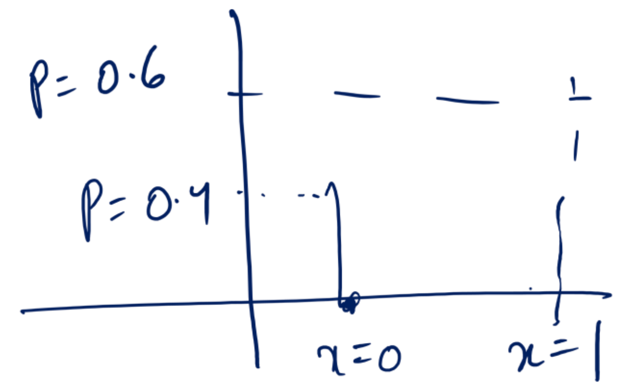
Discrete Probability Functions



- Examples:
  - Bernoulli Distribution:
    - $\begin{cases} 1-p & for\ x=0 \\ p & for\ x=1 \end{cases}$

In Bernoulli, just a **single** trial is conducted

  - Binomial Distribution:
    - $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$

k is number of successes

n-k is number of failures

$\binom{n}{k}$ The total number of ways of selection **k** distinct combinations of **n** trials, **irrespective of order**.

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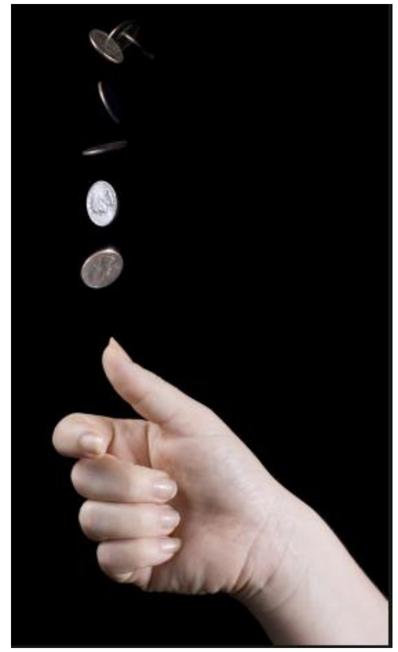
Example

X = Throw a dice



Y = Flip a coin

X and **Y** are random variables

N = total number of trials

$\boldsymbol{n_{ij}}$ = Number of occurrence

X

		$x_{i=1}=1$	$x_{i=2}=2$	$x_{i=3}=3$	$x_{i=4}=4$	$x_{i=5}=5$	$x_{i=6}=6$	C_j
Y	$y_{j=2}=tail$	$\boxed{n_{ij}=3}$	$n_{ij}=4$	$n_{ij}=2$	$n_{ij}=5$	$n_{ij}=1$	$n_{ij}=5$	20
	$y_{j=1}=head$	$n_{ij}=2$	$n_{ij}=2$	$n_{ij}=4$	$n_{ij}=2$	$n_{ij}=4$	$n_{ij}=1$	15
	C_i	5	6	6	7	5	6	N=35

X

		$x_{i=1}=1$	$x_{i=2}=2$	$x_{i=3}=3$	$x_{i=4}=4$	$x_{i=5}=5$	$x_{i=6}=6$	C_j
Y	$y_{j=2}=tail$	$\boxed{n_{ij}=3}$	$n_{ij}=4$	$n_{ij}=2$	$n_{ij}=5$	$n_{ij}=1$	$n_{ij}=5$	20
	$y_{j=1}=head$	$n_{ij}=2$	$n_{ij}=2$	$n_{ij}=4$	$n_{ij}=2$	$n_{ij}=4$	$n_{ij}=1$	15
	C_i	5	6	6	7	5	6	N=35

joint prob.

$$p(x=1, y=t) = \frac{3}{35} = \frac{n_{ij}}{N}$$

marginal prob

$$p(x=4) = \frac{(5+2)}{35} = \frac{C_i}{N}$$

$$p(y=t) = 20/35 = C_j/N$$

SUM RULE

$$p(x) = \sum_y p(x, y)$$

$$p(x=3 \mid y=t) = \frac{2}{20} = \frac{n_{ij}}{C_j}$$

$$p(y=t \mid x=3) = \frac{2}{6} = \frac{n_{ij}}{C_i}$$

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Recap

$p(x)$ or $p(x=a)$

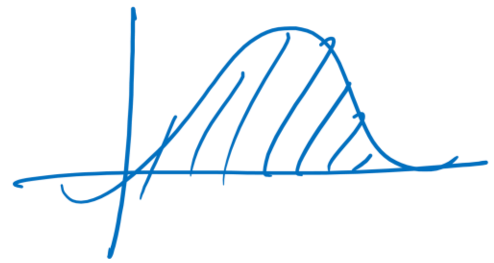
random variable

event

set of all possible outcomes

1 particular outcome

- Sample Space and Event
- Probability Distribution Function



continous → probability density function

discrete → probability mass function

- Objective Function

$\boxed{f(x)}$ has parameters $\underline{\theta}$

Gaussian dist. $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

$\{\mu, \sigma\} \in \theta$

- Joint Probability, Marginal Probability & Conditional Probability

$p(x, y)$

$p(a, b, c \ldots n)$

$p(x)$

$p(x \mid y)$

shrinking sample space

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Example

X = Throw a dice

Y = Flip a coin

X and **Y** are random variables

N = total number of trials

$\boldsymbol{n_{ij}}$ = Number of occurrence

X

Y		$x_{i=1} = 1$	$x_{i=2} = 2$	$x_{i=3} = 3$	$x_{i=4} = 4$	$x_{i=5} = 5$	$x_{i=6} = 6$	C_j
	$y_{j=2} = tail$	$n_{ij} = 3$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 5$	$n_{ij} = 1$	$n_{ij} = 5$	20
	$y_{j=1} = head$	$n_{ij} = 2$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 1$	15
	C_i	5	6	6	7	5	6	N=35

		X						
		$x_{i=1} = 1$	$x_{i=2} = 2$	$x_{i=3} = 3$	$x_{i=4} = 4$	$x_{i=5} = 5$	$x_{i=6} = 6$	C_j
Y	$y_{j=2} = tail$	$n_{ij} = 3$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 5$	$n_{ij} = 1$	$n_{ij} = 5$	20
	$y_{j=1} = head$	$n_{ij} = 2$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 1$	15
	C_i	5	6	6	7	5	6	N=35

$$p(y=t, x=4) = 5/35 = n_{ij}/N$$

$$p(y=t) = 20/35 = C_j/N = \boxed{\sum_x p(x,y) = p(y)}$$

summing over all X

SUM RULE

$$p(y=t \mid x=4) = \frac{5}{7} = \frac{n_{ij}}{C_i}$$

$$p(x=4 \mid y=t) = \frac{5}{20} = \frac{n_{ij}}{C_j}$$

x is a more informative r.v.

$$p(x \mid y) = \frac{n_{ij}}{C_j} = \frac{n_{ij}}{N} \cdot \frac{N}{C_j} = \frac{p(x,y)}{p(y)}$$

$$\Rightarrow \boxed{p(x \mid y)\, p(y) = p(x,y)}$$

PRODUCT RULE

$$\boxed{\begin{aligned} p(x,y) &= p(y \mid x)\, p(x) \\ &= p(x \mid y)\, p(y) \end{aligned}}$$

Probability: $$p(X = x_i) = \frac{c_i}{N}$$

Joint probability: $$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

Conditional probability: $$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

Do not like list

1. for loops
2. inverse
3. matrix multiplication
4. joint probability

Sum rule

$$p(X = x_i) = \sum_{j=1}^{L} p\left(X = x_i, Y = y_j\right) \Rightarrow \; p(X) = \sum_{Y} P(X, Y)$$

Product rule

$$p\left(X = x_i, Y = y_j\right) = \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i}\frac{c_i}{N} = p\left(Y = y_j \middle| X = x_i\right) p(X = x_i)$$

$$p(X, Y) = p(Y|X)p(X)$$

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Conditional Independence

$$P(h,f,v,d) = P(h|f,v,d)P(f,v,d)$$
$$= P(h|f,d)P(f|v,d)P(v,d)$$
$$= P(h|f,d)P(f|v)P(v|d)P(d)$$
$$= P(h|f,d)P(f|v)P(v)P(d)$$

- Examples:

$$P(Virus|\,DrinkBeer) = P(Virus)$$

iff Virus is independent of Drink Beer

$$P(Flu \mid \cancel{Virus}, DrinkBeer) = P(Flu \mid Virus)$$

iff Flu is independent of Drink Beer, given Virus

$$P(Headache \mid Flu, \cancel{Virus}, DrinkBeer)$$
$$= P(Headache|Flu, DrinkBeer)$$

iff Headache is independent of Virus, given Flu and Drink Beer

Assume the above independence, we obtain:

$$P(Headache, Flue, Virus, DrinkBeer)$$
$$= P(Headache|Flu, Virus, DrinkBeer)\ P(Flu|Virus, DrinkBeer)$$
$$\mathrm{P(Virus|DrinkBeer)P(DrinkBeer)}$$
$$= \mathrm{P(Headache|Flu, DrinkBeer)P(Flu|Virus)P(Virus)P(DrinkBeer)}$$

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Bayes' Rule

product rule

$p(x,y) = p(x|y) \cdot p(y)$

- P(X|Y)= Fraction of the worlds in which X is true given that Y is also true.

$$\boxed{p(x|y) = \frac{p(x,y)}{p(y)}}$$

- For example:
  - H="Having a headache"
  - F="Coming down with flu"
  - $P(Headche|Flu)$ = fraction of flu-inflicted worlds in which you have a headache. How to calculate?

$$p(x|y) = \frac{p(y|x)\,p(x)}{p(y)}$$

- Definition:

$$P(X|Y) = \frac{P(X,Y)}{P(Y)} = \frac{\mathrm{P(Y|X)P(X)}}{\mathrm{P(Y)}}$$

Corollary:

$$P(X,Y) = P(Y|X)P(X)$$

This is called Bayes Rule

Bayes' Rule

- $P(Headache|Flu) = \frac{P(Headache,Flu)}{P(Flu)}$

$$= \frac{P(Flu|Headache)P(Headache)}{P(Flu)}$$

when y is a binary variable

Other cases:

- $P(Y|X) = \frac{P(X|Y)P(Y)}{P(X|Y)P(Y)+P(X|\neg Y)P(\neg Y)}$

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)} = \frac{p(x|y)p(y)}{\sum_y p(x,y)} = \frac{p(x|y)p(y)}{p(x,y)+p(x,\neg y)} = \frac{p(x|y)p(y)}{p(x|y)p(y)+p(x|\neg y)p(\neg y)}$$

- $P(Y = y_i|X) = \frac{P(X|Y)P(Y)}{\sum_{i \in S} P(X|Y = y_i)P(Y=y_i)}$

- $P(Y|X,Z) = \frac{P(X|Y,Z)P(Y,Z)}{P(X,Z)} =$

$$\frac{P(X|Y,Z)P(Y,Z)}{P(X|Y,Z)P(Y,Z)+P(X|\neg Y,Z)P(\neg Y,Z)}$$

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In ML → Expectation or weighted average is the Same as arithmetic mean

Mean and Variance

- Expectation: The mean value, center of mass, first moment:

continuous

$$E_X[g(X)] = \int_{-\infty}^{\infty} g(x)p_X(x)dx = \mu$$

$E[g(x)]$

discrete

$E[g(x)] = \sum_{i=1}^{n} g(x_i)p(x_i)$

- N-th moment: $g(x) = x^n$
- N-th central moment: $g(x) = (x-\mu)^n$
- Mean: $E_X[X] = \int_{-\infty}^{\infty} xp_X(x)dx$
  - $E[\alpha X] = \alpha E[X]$
  - $E[\alpha + X] = \alpha + E[X]$
- Variance(Second central moment): $Var(x) = E_X[(X - E_X[X])^2] = E_X[X^2] - E_X[X]^2$
  - $Var(\alpha X) = \alpha^2 Var(X)$
  - $Var(\alpha + X) = Var(X)$

$E[x - E[x]]^2$

Expected value and average

biased 3 sided die

$x = [1, 2, 3]$ limited data

$p(x) = \left[\underbrace{\tfrac{1}{6}, \tfrac{1}{3}, \tfrac{1}{2}}_{\text{add up to 1.}}\right]$

valid pmf ??

$g(x) = x$

$E[g(x)] = \sum_{i=1}^{3} g(x_i)p(x_i)$

$= 1 \times 1/6 + 2 \times 1/3 + 3 \times 1/2 = \frac{14}{6}$

$\mu = \frac{1+2+3}{3} = 2$

We roll this biased die 6 times $x = [1, 2, 2, 3, 3, 3]$ $\mu = \frac{1+2+2+3+3+3}{6} = \frac{14}{6}$

Expectation determined by probability distribution. Arithmetic average determined by observed outcomes of trials.

Both are the same in our class because we assume we have sufficient data that models the distribution. So, the arithmetic average is a good estimate of the true expectation (Law of Large Numbers).

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Variance and average:

how much our data varies from the mean

height

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{n \times d}$$

3 $\to n$, $d \leftarrow 1$

for loop

$$\text{Variance} = \frac{1}{N} \sum_{i=1}^{n} (x-\mu)^2$$

$$= E[x - \mu]^2 = E[x - E[x]]^2$$

mean

scalar

$$\text{Var} = \frac{1}{n} \bar{X}^T \bar{X}$$

$\bar{X}$ is the centered matrix

$\bar{X}_{3\times1}$

$$\bar{X} = \begin{bmatrix} 1-\mu_n \\ 2-\mu_n \\ 3-\mu_n \end{bmatrix} \qquad \frac{1}{N} \bar{X}^T_{1\times3} \bar{X}$$

$$\text{Var} = \frac{1}{N} \begin{bmatrix} 1-\mu_n & 2-\mu_n & 3-\mu_n \end{bmatrix} \begin{bmatrix} 1-\mu_n \\ 2-\mu_n \\ 3-\mu_n \end{bmatrix}$$

$$\text{Var} = \frac{1}{N}\left((1-\mu_n)^2 + (2-\mu_n)^2 + (3-\mu_n)^2\right)$$

1×1 scalar because we have only 1 feature.

If d features, $\text{Var}_{d\times d} \to$ Covariance matrix

Covariance:

If we have d features, we can compute covariance matrix of size $d \times d$.

$$Cov(\Sigma) = \frac{1}{N}\bar{X}^T \bar{X}$$

$$X = \begin{array}{c} \begin{array}{cc} h & w \end{array} \\ \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{n \times d} \end{array}$$

3 2

$$\bar{X} = \begin{bmatrix} 1-\mu_h & 4-\mu_w \\ 2-\mu_h & 5-\mu_w \\ 3-\mu_h & 6-\mu_w \end{bmatrix}$$

$$\frac{1}{N}\begin{bmatrix} 1-\mu_h & 2-\mu_h & 3-\mu_h \\ 4-\mu_w & 5-\mu_w & 6-\mu_w \end{bmatrix}\begin{bmatrix} 1-\mu_h & 4-\mu_w \\ 2-\mu_h & 5-\mu_w \\ 3-\mu_h & 6-\mu_w \end{bmatrix}$$

1st term: $(1-\mu_h)^2 + (2-\mu_h)^2 + (3-\mu_h)^2$

4th term: $(4-\mu_w)^2 + (5-\mu_w)^2 + (6-\mu_w)^2$

(2x2)

$$Cov = \begin{array}{cc} \begin{array}{cc} h & w \end{array} & \\ \begin{bmatrix} \sigma_h^2 & \sigma_{hw}^2 \\ \sigma_{wh}^2 & \sigma_w^2 \end{bmatrix} & \begin{array}{c} h \\ w \end{array} \end{array}$$

high no. high no.

$$\sigma_{wh} = (1-\mu_h)(4-\mu_w) + \ldots\ldots$$

$\sigma_{wh} = \sigma_{hw}$? YES Cov → Symmetrical

σ_{wh} tells us how weight & height correlate

$\sigma_{wh} = 0$ $\sigma_{wh} > 0$

Correlation:

$$Cov = \frac{1}{N}\bar{X}^T \bar{X}$$

$$Cor = \frac{1}{N}\bar{X}^{*T}\bar{X}^*$$

Normalize our features

$\to [0,1]$ ✓ ✗

$\underbrace{[-1,1]}_{\text{standardization}}$

$\sigma^2 \leftarrow$ variance

$\sigma \rightarrow$ std. dev

$$\bar{X}^* = \frac{\bar{X}}{\sigma}$$

$$\bar{X}^* = \begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & \frac{4-\mu_w}{\sigma_w} \\ \frac{2-\mu_h}{\sigma_h} & \frac{5-\mu_w}{\sigma_w} \\ \frac{3-\mu_h}{\sigma_h} & \frac{6-\mu_w}{\sigma_w} \end{bmatrix}$$

$$Cor[1,1] = \frac{1}{N}\begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & \frac{2-\mu_h}{\sigma_h} & \frac{3-\mu_h}{\sigma_h} \\ \frac{4-\mu_w}{\sigma_w} & \frac{5-\mu_w}{\sigma_w} & \frac{6-\mu_w}{\sigma_w} \end{bmatrix}\begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & 4-\mu_w \\ \frac{2-\mu_h}{\sigma_h} & \\ \frac{3-\mu_h}{\sigma_h} & \end{bmatrix}$$

$$\frac{1}{N}\left(\frac{(1-\mu_h)^2}{\sigma_h^2} + \frac{(2-\mu_h)^2}{\sigma_h^2} + \frac{(3-\mu_h)^2}{\sigma_h^2}\right) = \frac{\sigma_h^2}{\sigma_h^2} = 1$$

$$Cor = \begin{array}{c} \\ \end{array}\begin{array}{cc} h & w \end{array}$$

$$Cor = \begin{bmatrix} 1 & -1<[\ ]<1 \\ -1<[\ ]<1 & 1 \end{bmatrix} \begin{array}{c} h \\ w \end{array}$$

Useful in EDA. If features are correlated, then data may have redundancy.

Uncorrelated vs Independent RV

Uncorrelated (Definition: Cov(X,Y)=0.)
•Means no **linear relationship** between X and Y.
•Does **not** rule out non-linear dependence.
•Example: X∼U(−1,1), Y=x^2 → Uncorrelated but dependent.

Independent (Definition: P(X∈A,Y∈B)=P(X∈A)P(Y∈B))
•Stronger condition: knowing one variable gives **no information** about the other.
•Independence ⇒ Uncorrelated (if variances exist).

Key Differences
•Independence is a **stronger** property.
•Uncorrelated only removes **linear relationships**.
•Uncorrelated ⇏ Independent.

ML Relevance

- Most ML models care about uncorrelatedness because they only model linear relationships.
- True independence is rarer and much harder to check, but it's what we assume in stronger probabilistic models

For Joint Distributions

- Expectation and Covariance:
  - $E[X + Y] = E[X] + E[Y]$
  - $cov(X, Y) = E[(X - E_X[X])(Y - E_Y(Y)] = \boxed{E[XY] - E[X]E[Y]}$
  - $Var(X + Y) = Var(X) + 2cov(X, Y) + Var(Y)$

Recap

- Expectation and arithmetic mean
- Variance
- Covariance
- Correlation
- Uncorrelated vs Independent r.v.

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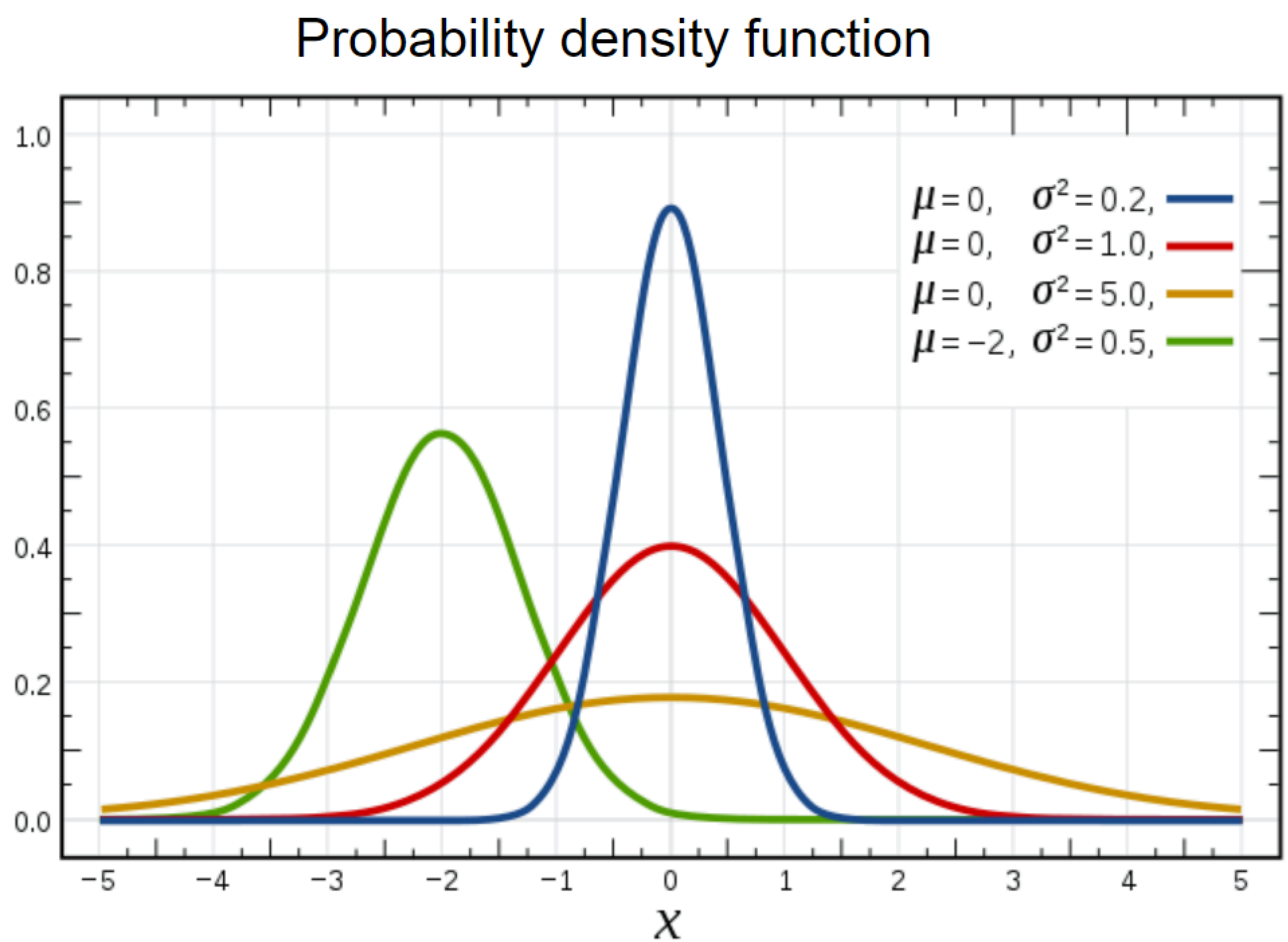
Gaussian Distribution

- Gaussian Distribution: $$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



[Probability versus likelihood]

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Prob vs Likelihood

Prob vs Likelihood

Multivariate Gaussian Distribution

$$p(x|\mu,\Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x-\mu)^{\top}\Sigma^{-1}(x-\mu)\}$$

- Moment Parameterization $\mu = E(X)$

$$\Sigma = Cov(X) = E[(X-\mu)(X-\mu)^{\top}]$$

- Mahalanobis Distance $\Delta^2 = (x-\mu)^{\top}\Sigma^{-1}(x-\mu)$

- Tons of applications (MoG, FA, PPCA, Kalman filter,...)

Properties of Gaussian Distribution

- The linear transform of a Gaussian r.v. is a Gaussian. Remember that no matter how x is distributed

$$E(AX+b) = AE(X) + b$$
$$Cov(AX+b) = ACov(X)A^\top$$

this means that for Gaussian distributed quantities:

$$X \sim N(\mu, \Sigma) \;\; \rightarrow \;\; AX + b \sim N(A\mu + b, A\Sigma A^\top)$$

- The sum of two independent Gaussian r.v. is a Gaussian

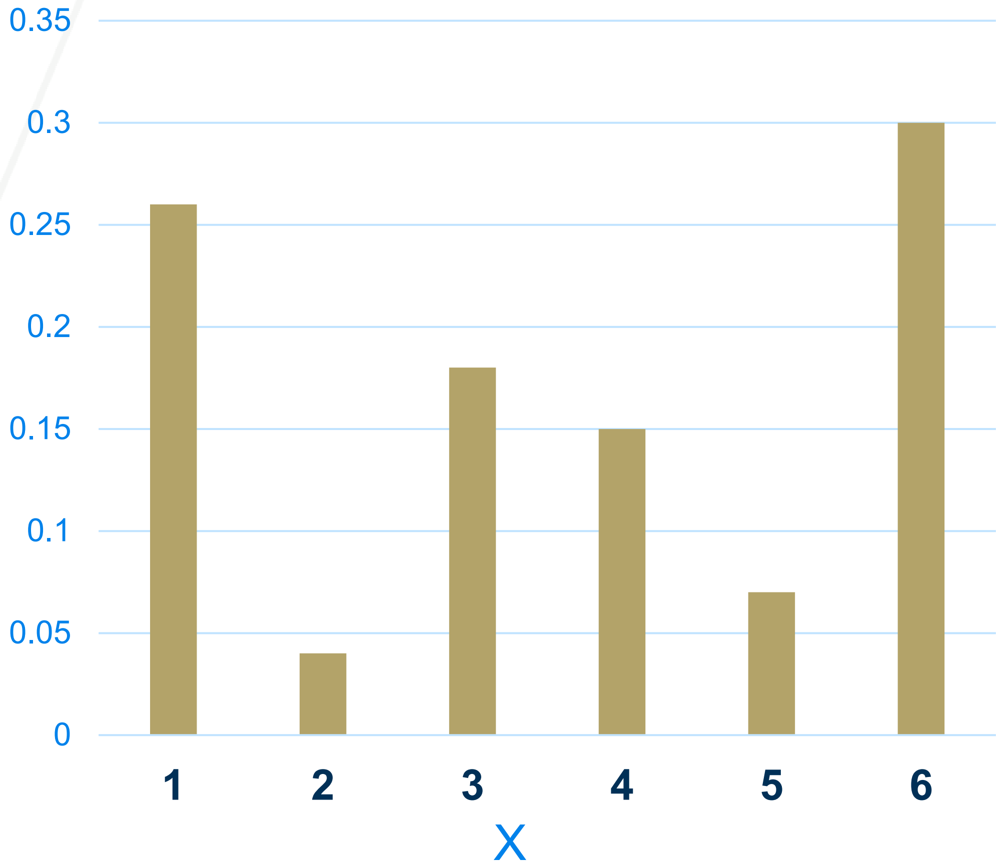
$$\mathrm{Y} = \mathrm{X}_1 + \mathrm{X}_2,\; \mathrm{X}_1 \perp X_2 \;\rightarrow \mu_y = \mu_1 + \mu_2, \Sigma_y = \Sigma_1 + \Sigma_2$$

- The multiplication of two Gaussian functions is another Gaussian function (although no longer normalized)

$$N(a, A)N(b, B) \propto N(c, C),$$
$$where\ C = (A^{-1} + B^{-1})^{-1}, c = \mathrm{C}A^{-1}a + CB^{-1}b$$

Central Limit Theorem

Probability mass function of a **biased** dice



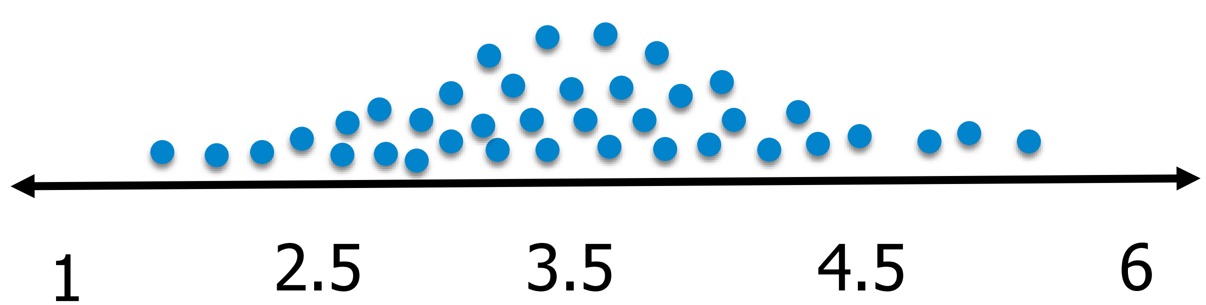
Let's say, I am going to get a sample from this pmf having a size of $\boldsymbol{n = 4}$

$$S_1 = \{1,1,1,6\} \Rightarrow E(S_1) = 2.25$$

$$S_2 = \{1,1,3,6\} \Rightarrow E(S_2) = 2.75$$

$$\vdots$$

$$S_m = \{1,4,6,6\} \Rightarrow E(S_m) = 4.25$$



According to CLT, it will follow a bell curve distribution (normal distribution)

Outline

- Probability Distributions
- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation

Maximum Likelihood Estimation

- Probability: inferring probabilistic quantities for data given fixed models (e.g. prob. of events, marginals, conditionals, etc).
- Statistics: inferring a model given fixed data observations (e.g. clustering, classification, regression).

Main assumption:

Independent and identically distributed random variables

i.i.d

Maximum Likelihood Estimation

For Bernoulli (i.e. flip a coin):

Objective function: $P(x_i|\theta) = \theta^{x_i}(1-\theta)^{1-x_i}$ $\quad x_i \in \{0,1\}\ or\ \{head, tail\}$

$$L(\theta|X) = L(\theta|X = x_1, X = x_2, X = x_3, \ldots, X = x_n)$$

i.i.d assumption

$$L(\theta|X) = \prod_{i=1}^{n} P(x_i|\theta)$$

$$L(\theta|X) = \prod_{i=1}^{n} P(x_i|\theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}$$

$$L(\theta|X) = \theta^{x_1}(1-\theta)^{1-x_1} \times \theta^{x_2}(1-\theta)^{1-x_2} \ldots \times \theta^{x_n}(1-\theta)^{1-x_n} =$$
$$= \theta^{\sum x_i}(1-\theta)^{\sum(1-x_i)}$$

We don't like multiplication, let's convert it into summation

What's the trick? Take the log

$$L(\theta|X) = \theta^{\sum x_i}(1-\theta)^{\sum(1-x_i)}$$

$$logL(\theta|X) = l(\theta|X) = \log(\theta)\sum_{i=1}^{n} x_i + \log(1-\theta)\sum_{i=1}^{n}(1-x_i)$$

How to optimize θ?

$$\frac{\partial l(\theta|X)}{\partial\theta} = 0 \qquad \frac{\sum_{i=1}^{n} x_i}{\theta} - \frac{\sum_{i=1}^{n}(1-x_i)}{1-\theta} = 0$$

$$\theta = \frac{1}{n}\sum_{i=1}^{n} x_i$$

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